Tenghan Zhong | tenghanz@usc.edu | +1 2135072381
https://tenghanzhong.github.io/

Academic Background

University of Southern California (USC) Sep.2024 - May.2026
Master of Science in Mathematical Finance, USC Dornsife College Los Angeles, CA
GPA: 4.0/4.0

Nanchang University - Project 211 Sep.2020 - Jul.2024
Bachelor of Economics in Mathematical Finance, School of Mathematics and Computer Science Nan Chang, CN
GPA: 3.62/4.0; 87.02/100

Graduate-Level Courses

✓ Stochastic Calculus and Mathematical Finance A & B
✓ Mathematical Statistics (PhD Level)
✓ Linear Programming (PhD Level)
✓ Machine Learning in Quantitative Finance
✓ Financial Informatics and Simulation
✓ Time Series Analysis
✓ Big Data Econometrics
✓ Investment Analysis and Portfolio Management

Research Papers

"Marking-Aware Sequential VaR Recalibration for Standardized Option Books"
Tenghan Zhong and Keyuan Wu - Accepted for long presentation, IEEE CIFEr 2026. [arXiv]

"Risk-Sensitive Specialist Routing for Volatility Forecasting"
Tenghan Zhong — Accepted as a Regular Paper (Oral Presentation) at IEEE SMC 2026. [arXiv]

"Non-Spanning Identification of Scheduled Event Risk in Option Pricing"
Tenghan Zhong — Under review at ACM ICAIF 2026. [arXiv]

"Quantifying Repair Risk in Crypto Option Surfaces"
Working paper.

Research Experience

Research Assistant -- Quantitative Research Group
Supervisor: Guang Cheng, Prof. of Statistics and Data Science, Graduate Vice Chair (UCLA) Jun.2025-Sep.2025
✓ Engineered an automated data pipeline using Python scripts to continuously extract, transform, and load over 5 million minute-level records from Polymarket and Deribit APIs into a PostgreSQL database, which served as the foundation for subsequent empirical and methodological analysis.
✓ Developed a cross-market mispricing detection framework that synthetically replicates Polymarket interval payoffs via dynamically selected call/put vertical spreads on Deribit; backtested more than 200 trades over three months under realistic transaction costs, documenting robust, positive, low-volatility returns.
✓ Built an in-play microstructure-based decision model for MLB markets on Polymarket; optimized it by performing feature engineering on game-state and microstructure data. The optimized model roughly doubled risk-adjusted performance and halved maximum drawdown, and showed consistent positive returns in live deployment.

Undergraduate Thesis -- Dynamic Pricing of ETF Options Based on Recurrent Neural Networks
--- github.com/TenghanZhong/GRU-Heston
Supervisor: Lidan Liao, Professor of Mathematics(Nanchang University) Dec.2023-Feb.2024
- ✓ Proposed a structural-residual pricing framework that integrates Heston pricing with GRU/LSTM residual correction, reducing out-of-sample pricing MSE and MAPE by at least 70% relative to structural-model benchmarks.
- ✓ Developed a rolling Heston calibration and recurrent residual-correction pipeline using 2.0M+ one-minute ETF-option observations.
- ✓ Validated the residual-learning design using statistical dependence tests and rolling out-of-sample evaluation, confirming persistent temporal structure in Heston pricing residuals.

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Project Experience

Calibration of Power-Type Derivatives for Rough Volatility with Jumps
---*github.com/TenghanZhong/GPU-NN-Option-Calibration*
Supervisor: Weixuan Xia, Assistant Professor of Mathematics (USC) Feb.2025-May.2025

- ✓ Developed a GPU+FNN-accelerated two-stage calibration (genetic algorithm followed by L-BFGS-B) for a rough-OU volatility model with tempered-stable jumps; reduced runtime from over 24 hours to under 20 minutes (~98%) and improved fit accuracy by 30% (MAE 0.047→0.033).
- ✓ Used high-frequency activity-index estimates to identify the pure-jump regime and impose data-driven parameter constraints, improving calibration stability across maturities.

Hybrid Approaches to Crypto Volatility Prediction Using GARCH and GRU Model
--- github.com/TenghanZhong/Crypto_Volatility_prediction
Supervisor: Ratika Narag, Professor of Economics (USC) Sep.2024-Dec.2024

- ✓ Developed two types of hybrid architectures (GRU-GARCH, GRU-GARCH-Error) that combine GARCH-family volatility structure with recurrent learning for multi-horizon crypto-volatility forecasting.
- ✓ Evaluated the models on 830,000+ five-minute Coinbase observations using rolling forecasts, with GRU-AVARCH outperforming GARCH-family benchmarks by at least 15.0% and standalone GRU by at least 11.7% across T+1 to T+6 horizons.
- ✓ Derived model-specific convexity and Hessian conditions for GARCH, GJR-GARCH, EGARCH, and AVARCH, linking likelihood conditioning to empirical convergence behavior and model selection.

Working Experience

USC Dornsife College of Letters, Arts and Sciences. / *TA* Sep.2025-May.2026 / Los Angeles, CA

- ✓ Department of Mathematics: MATH 125g: Calculus I (Fall 2025)
- ✓ Department of Mathematics: MATH 126: Calculus II (Spring 2026)

Vanho Securities Co. Ltd./*Asset Management Department* Sep.2023-Nov.2023 / Shenzhen, CN

- ✓ Reproduced and extended quantitative timing and value-stock-selection strategies based on Everbright Securities and Huatai Securities research, using Tushare market data and Python.
- ✓ Refined strategy parameters through sensitivity analysis and additional constraints, increasing backtested excess returns by approximately 200% relative to the reproduced baseline while reducing maximum drawdown by 20%.
- ✓ Built a unified backtesting framework for timing and stock-selection strategies, incorporating performance attribution, risk metrics, visualization, and statistical robustness checks.

Ping An Securities Co., Ltd./*Brokerage Division* Jul.2022-Aug.2022 / Shenzhen, CN

- ✓ Processed and validated private-equity income and product-level financial data using Excel and Python, supporting periodic performance monitoring and reporting.
- ✓ Prepared weekly and monthly reports on private-equity products, analyzing operating performance, return trends, and key portfolio metrics.

Skills and Certifications

- ✓ Programming: Python, R, SQL
- ✓ Tools: Git/GitHub, PostgreSQL, LaTeX, Microsoft Office
- ✓ Languages: Chinese (native), English (proficient)
- ✓ Honors and Awards:
  - 2023.09 First-Class Scholarship
  - 2023.03 Outstanding Student Cadres
  - 2022.09 First-Class Scholarship